LEAKLISTEN · PROJECT PLAN

LeakListen project plan

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/leaklisten		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-01	Amish Chadha	First plan at stage P3 (kit 1.9.1 upgrade); P4 to P6 shown as planned work on hold

LeakListen project plan

LeakListen is at stage P3 (design outputs), TRL 3. The calculation note, the D3 model of all three options, the drawing, the priced BOM and the build plan exist. The next gate, P4 (verification, TRL 4), is on hold while the portfolio stage cap is P3. The decision requested is the concept choice (LKL-DDR-004). Method: STANDARDS sections 19 and 23.

Workstreams

Table 1. Workstreams.

Workstream	Owner (role)	Deliverable	Status
Concept choice	Amish Chadha	LKL-DEC-001 D10, LKL-DDR-004	Proposed, awaiting Amish
Partner and site	Amish Chadha	A UK water company with metallic mains and district metered areas (LKL-DEC-001, 2026-10-02)	Not started
Requirements	Design lead	LKL-URS-001, LKL-PRS-001, LKL-RTM-001	Done on paper; partner review pending
Calculations and model	Design lead	LKL-CAL-001; D3 model of options A, B and C; STEP and STL	Done 2026-11-01
Drawings, BOM, build plan	Design lead	LKL-DWG-001, DWG-101 to 109, bom/bom.csv, LKL-BLD-001	Done for option A
Risk, market, regulatory	Design lead	LKL-RSK-001 with design FMEA, LKL-MKT-001, LKL-REG-001	Done 2026-11-01
Renders	Coordinator	Options picture and product renders (Blender, on a Mac)	Queued
Verification (P4)	Partner and lab	Cover loss, shaker, noise, cell pulse and magnet pull tests	On hold (stage cap P3)

Resources needed

Table 2. Resources.

Resource	Deliverable	Effort	Estimated cost (USD)
Design lead (Claude with Amish's review)	P1 to P3 documents	2 sessions (done)	0
Prototype parts	One logger set (P4, on hold)		142 (bom/bom.csv)
Main board design	Board for the STM32WL core (P4, on hold)	To assess	To assess
Gateway	A LoRaWAN gateway near the pilot district (the lab's TwinKit)	To assess	To assess
Partner site survey	Cover types, neck sizes, pipe materials	1 to 2 days	To assess

Stage gates

Table 3. Stages (P4 to P6 on hold while the stage cap is P3).

Stage	Gate criteria	Status
P0 Proposal	PRB, STM, OPT, REG, FEA, PLN	Done 2026-11-01 (written in the kit 1.9.1 upgrade)
P1 Definition	URS, PRS draft, RSK, MKT, REG complete	Done 2026-11-01
P2 Design inputs	PRS targets, CAL, RTM, DDR, STEM check closed by calculation	Done 2026-11-01; 13 STEM rows need field data or datasheets
P3 Design outputs	Model, STEP, DWG, BOM, BLD, DEC, FMEA	Done 2026-11-01
P4 Verification	Cover loss, resonance and noise measurements, build	On hold
P5 to P6	Field trial with a partner, launch	On hold

Next steps and decision requested

Requested: Amish's concept choice (LKL-DDR-004) and his view on the optional seismic mass change (LKL-DEC-001 D12). The recommended next step is a partner and, when the cap allows, TRL 4: the cover loss measurement and a shaker comparison of the sensor.

References

Ofcom. "IR 2030 UK Interface Requirements 2030." Accessed November 1, 2026. https://www.ofcom.org.uk/__data/assets/pdf_file/0028/84970/ir-2030.pdf.

SubSurface Leak Detection. "SubSurface Enigma 3 Hybrid Logger System." Tiger Supplies. Accessed November 1, 2026. https://tigersupplies.com/products/subsurface-enigma-3-hybrid-logger-system.